Week 7: Introduction to Statistics for Data Science

Loading the data.

In [2]:
```
import pandas as pd
import numpy as np

df = pd.read_csv('business_data.csv')

df.head()
```

Out[2]:

	Date	Store_Region	Promo_Strategy	Daily_Visitors	Marketing_Spend	Daily_Revenue
0	2025-01-01	North	Standard	4856	1145.32	9876.45
1	2025-01-02	South	Aggressive_Discount	5234	1389.76	12543.21
2	2025-01-03	East	Standard	4923	1087.54	9234.67
3	2025-01-04	West	Aggressive_Discount	5512	1456.89	13012.34
4	2025-01-05	North	Standard	4789	1234.12	10456.78

descriptive statistics

In [ ]:
```
# shows all the descriptive statistics of all the numeric data column in dataset. Exe
desc_stats= df.describe()

#Median of all numerical data column
medians = df[['Daily_Visitors','Marketing_Spend','Daily_Revenue','Customer_Satisfacti

#mode for all categorical data column
mode_reg = df['Store_Region'].mode()[0]
mode_promo = df['Promo_Strategy'].mode()[0]
```

In [38]:
```
print("Descriptive statstics:")
print('='*50)
desc_stats
```

```
Descriptive statstics:
==================================================
```

Out[38]:

	Daily_Visitors	Marketing_Spend	Daily_Revenue	Customer_Satisfaction_Score
count	200.000000	200.000000	200.000000	200.000000
mean	5198.810000	1328.221850	11730.125300	4.254900
std	320.571998	169.174615	1837.564176	0.201831
min	4678.000000	1045.670000	8934.560000	3.840000
25%	4912.000000	1188.835000	10123.450000	4.097500
50%	5128.500000	1289.340000	11023.450000	4.240000
75%	5514.750000	1498.980000	13678.900000	4.430000
max	5745.000000	1634.560000	15123.450000	4.610000

In [39]:

```
print('='*50)
print("Median of all Numerical Data columns:")
print('='*40)
print(medians)
print('='*40)
print("Mode for categorical data coclumns")
print('='*40)
print(f"Store_Region:   {mode_reg}")
print(f"Promo_Strategy: {mode_promo}")
print('='*50)
```

```
==================================================
Median of all Numerical Data columns:
========================================
Daily_Visitors                  5128.50
Marketing_Spend                 1289.34
Daily_Revenue                  11023.45
Customer_Satisfaction_Score        4.24
dtype: float64
========================================
Mode for categorical data coclumns
========================================
Store_Region:   East
Promo_Strategy: Standard
==================================================
```

Statistical Analysis:

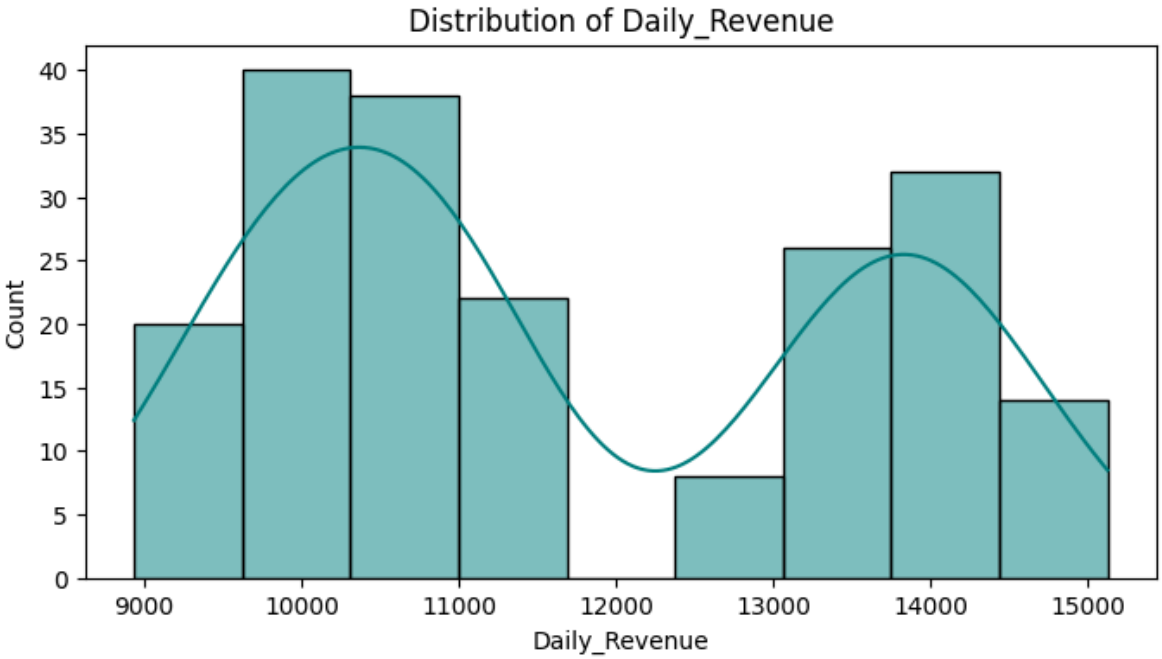
- 'Daily Revenue':
  - Mean = 11730.12
  - Median = 11023.45
  - difference is around 700, so there is outliers, In our case there is No mega sales day happened.
- 'Marketing Spend'
  - standard deviation = 169.17
  - comparing to its mean(which is 1328), our marketing spend is reasonable, a its fine.
- 'Customer_Satisfaction_Score'
  - the mean is 4.25. which is a very good thing.
  - It means every 4.25 people out of 5 people are statisfied with our business service.

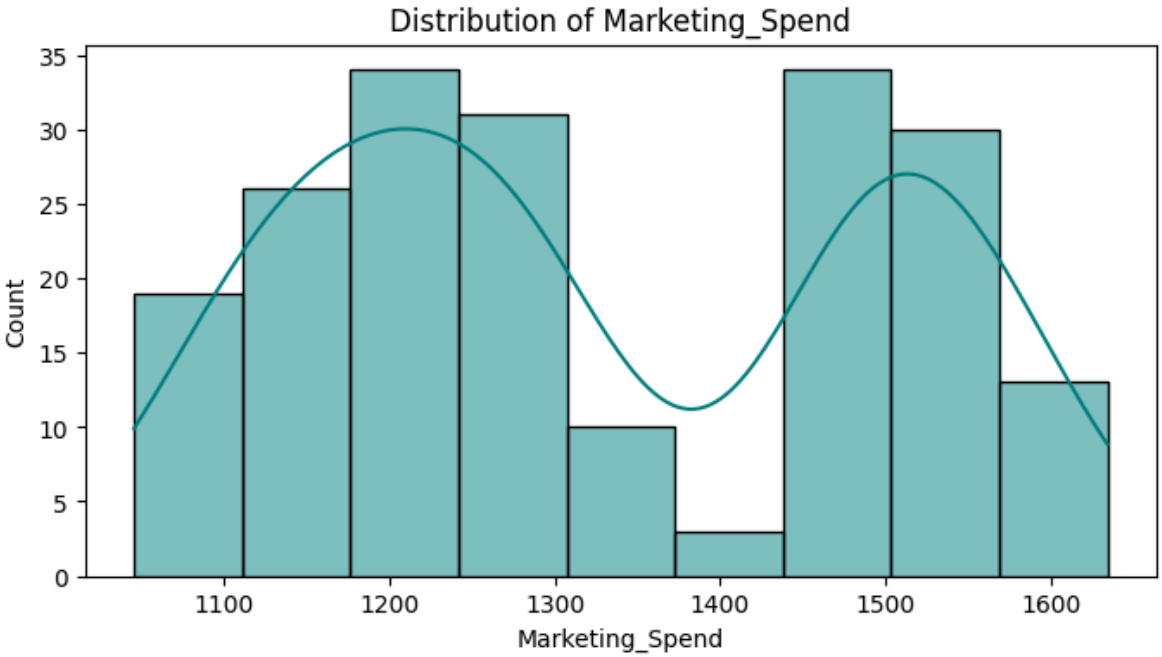
Data Distribution Analysis

In [5]:
```
import seaborn as sns
import matplotlib.pyplot as plt

cols_to_check = ['Daily_Revenue', 'Marketing_Spend']

for col in cols_to_check:
    plt.figure(figsize=(8, 4))
    sns.histplot(df[col], kde=True, color='teal')
    plt.title(f'Distribution of {col}')
    plt.show()
```





We have a 'M' shaped distribution which means we have 2 different peaks, all thanks to the promo_statergy. I guess the promo Stategy worked.

Shapiro_Wilk test is a most popular test for checking Normality amoung data.

In [6]:
```
from scipy.stats import shapiro

for col in cols_to_check:
    stat, p = shapiro(df[col])
    print(f'--- {col} ---')
    print(f'Statistics={stat:.3f}, p-value={p:.3f}')

    if p > 0.05:
        print("Result: Looks Gaussian/Normal (Fail to reject H0)")
    else:
        print("Result: Does not look Normal (Reject H0)")
    print("="*20)
```

```
--- Daily_Revenue ---
Statistics=0.891, p-value=0.000
Result: Does not look Normal (Reject H0)
====================
--- Marketing_Spend ---
Statistics=0.916, p-value=0.000
Result: Does not look Normal (Reject H0)
====================
```

It is as clear as possible, that our data is not normal. cuz of the p-values is strictly '0'

In [7]:
```
# Creating a new column with the log of Revenue
df['Log_Revenue'] = np.log(df['Daily_Revenue'])

# Now, run the Shapiro test on the NEW column
stat, p = shapiro(df['Log_Revenue'])
print(f"Log Revenue P-value: {p:.3f}")
```

```
Log Revenue P-value: 0.000
```

I tried to transform the daily revenue column with log() and create a new column.

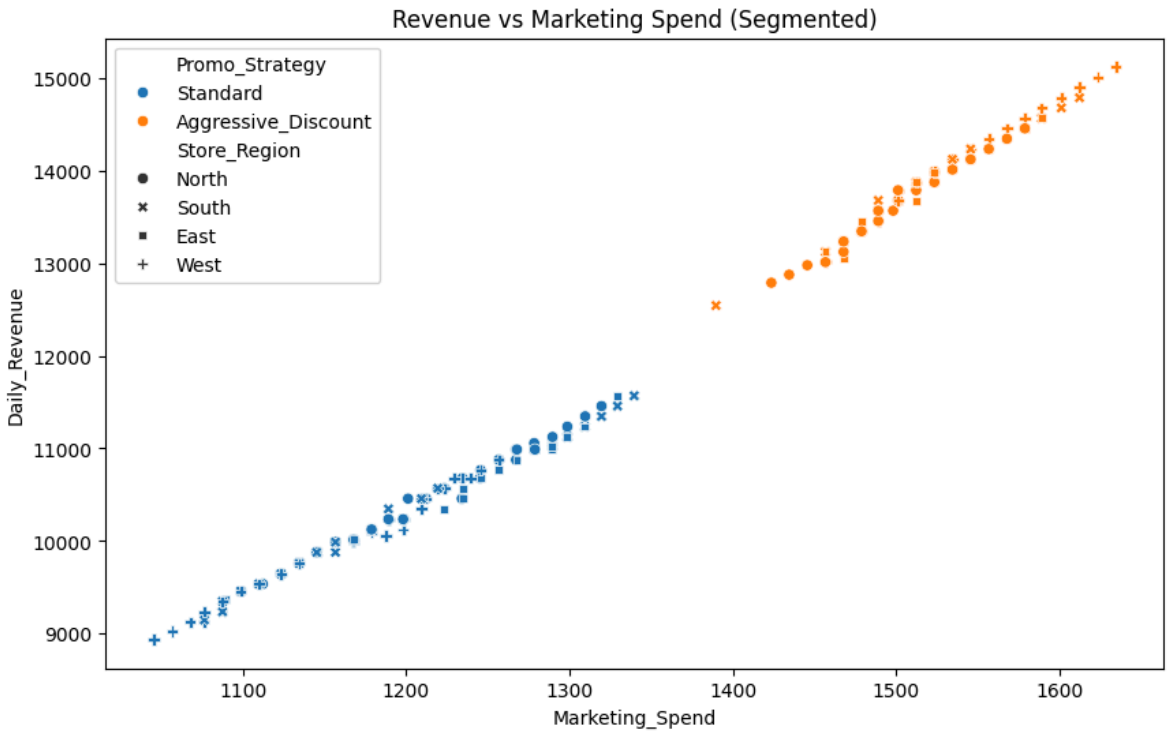
- the log() function will transforms the data in such a way that the column should look normal
- after transformation, the p-value remains '0'. so we quit to froceing the data to be normal and use different way to work-around.

Correlation Analysis

In [8]:
```
# Using 'spearman' for non-normal data
correlation = df['Marketing_Spend'].corr(df['Daily_Revenue'], method='spearman')
print(f"Spearman Correlation: {correlation:.3f}")
```

```
Spearman Correlation: 0.999
```

In [9]:
```
# This plot will show us if the "M" is caused by the Promo Strategy
plt.figure(figsize=(10, 6))
sns.scatterplot(data=df, x='Marketing_Spend', y='Daily_Revenue', hue='Promo_Strategy'
plt.title("Revenue vs Marketing Spend (Segmented)")
plt.show()
```



We can clear see that the promo_statergy used is a imapctful.

we will calcualate correlation for both group.

In [11]:
```
standard_stgy = df[df['Promo_Strategy']== 'Standard'][['Marketing_Spend','Daily_Rever
argDis_stgy = df[df['Promo_Strategy']== 'Aggressive_Discount'][['Marketing_Spend','Da

standard_stgy_corr = standard_stgy['Marketing_Spend'].corr(standard_stgy['Daily_Rever
argDis_stgy_corr = argDis_stgy['Marketing_Spend'].corr(argDis_stgy['Daily_Revenue'])

print(f'Correlation of Standard Statergy: {standard_stgy_corr:.4f}')
print(f'Correlation of Aggressive Discount Statergy: {argDis_stgy_corr:.4f}')
```

```
Correlation of Standard Statergy: 0.9957
Correlation of Aggressive Discount Statergy: 0.9907
```

By this we cal tell that the more the marketing spend the more the daily revenue

Hypothesis Testing

- Null hypothesis(h0): There is no real difference between the gap, it just a random gap
- Alternative Hypothesis(h1): The 'Aggressive_Discount' line is significantly higher than the 'Standard' line.

In [35]:
```
from scipy.stats import mannwhitneyu

stat, p = mannwhitneyu(standard_stgy['Daily_Revenue'], argDis_stgy['Daily_Revenue'])
print(f"Mann-Whitney U P-value: {p}")

if p < 0.05:
  print("We Reject the Null Hypothesis, and Accept the Alternative Hypothesis")
else:
  print("We failed to reject  the NULL hypothesis")

print(f'U Statistic is: {stat}')
```

```
Mann-Whitney U P-value: 4.975700200680058e-33
We Reject the Null Hypothesis, and Accept the Alternative Hypothesis
U Statistic is: 0.0
```

By this We can Confirm that the Aggressive Discount statergy is ligit and it is Impactful for the business.

Confidence Intervals

In [18]:
```
import scipy.stats as stats

for stg in ['Standard', 'Aggressive_Discount']:
  data = df[df['Promo_Strategy'] == stg]['Daily_Revenue']

  mean= np.mean(data)
  sem = stats.sem(data)   # standard error
  #calculating 95% confidence Intervals
  ci = stats.t.interval(0.95, len(data)-1, loc=mean, scale = sem)

  print(f'-----{stg}-----')
  print(f'Average Revenue: ₹{mean:.2f}')
  print(f'95% Confidence Intervals : ₹{ci[0]:.2f} to ₹{ci[1]:.2f}')
  print(f'Margin of error :± ₹{(ci[1]-mean):.2f}\n')
```

```
-----Standard-----
Average Revenue: ₹10323.15
95% Confidence Intervals : ₹10202.56 to ₹10443.75
Margin of error :± ₹120.60

-----Aggressive_Discount-----
Average Revenue: ₹13840.59
95% Confidence Intervals : ₹13715.12 to ₹13966.05
Margin of error :± ₹125.46
```

Aggressive Discount strategy shows a significantly higher average revenue than Standard strategy. The non-overlapping confidence intervals indicate that the difference is statistically meaningful, and the small margins of error suggest precise estimation.

simply put: Aggressive Discount performs better in revenue generation

Regression Analysis.

I will build a Multiple Linear regression model to predict the Daily_revenue

In [19]:
```
df1 = df.copy(deep=True) #a deep copy of original data.
```

In [24]:
```
#transforming data
#converting Promo_Statergy column from categorical to numerical

df1['Promo_encoded'] = df1['Promo_Strategy'].map({'Standard': 0, 'Aggressive_Discount
df1.head()
```

Out[24]:

	Date	Store_Region	Promo_Strategy	Daily_Visitors	Marketing_Spend	Daily_Revenue
0	2025-01-01	North	Standard	4856	1145.32	9876.45
1	2025-01-02	South	Aggressive_Discount	5234	1389.76	12543.21
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3	2025-01-04	West	Aggressive_Discount	5512	1456.89	13012.34
4	2025-01-05	North	Standard	4789	1234.12	10456.78

In [ ]:

```
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import r2_score

x = df1[['Marketing_Spend','Promo_encoded']]
y = df1['Daily_Revenue']

#Spliting the data into training and testing data
xtrain,xtest,ytrain,ytest = train_test_split(x,y,test_size=0.2, random_state=2, strat

#LinearRegression Model
lin = LinearRegression()

#Trianing the model
lin.fit(xtrain,ytrain)
```

Out[ ]:

```
▾ LinearRegression
▸ Parameters
```

In [ ]:

```
from sklearn.metrics import r2_score

#testing the model and printing it r2_score
ypred = lin.predict(xtest)

r2score = r2_score(ytest, ypred)

print(f'{r2score:.2%}')
```

```
99.74%
```

Our regression model predicted 99.76% of the data right, and remaining are causd by noise.

In [34]:

```
print(f"Intercept: {lin.intercept_:.2f}")
print(f"Coefficients: {lin.coef_}")
```

```
Intercept: -995.86
Coefficients: [  9.42168988 529.60959444]
```

- *the coefficient 1*: 9.421, tells us that if we spend 1 rupee we will get 9.42 rupee as revenue.

- *the coefficient 2*: 529.6, tells us that if we use any statergy we can gain 529 rupees by spending 1 rupee.

I want to make a simple prediction with some values, this is like a real_world deployment kind.

In [50]:
```
import warnings
warnings.filterwarnings('ignore')
```

You can change the new_spend and new_promo(selecting either Standard(0) or Aggressive_discount(1))

In [51]:
```
new_spend = 1000  # What we plan to spend
new_promo = 1     # 1 for Aggressive_Discount, 0 for Standard Statergy

# 2. Reshape for the model (it expects a 2D array/list of lists)
new_data = [[new_spend, new_promo]]

# 3. Get the prediction
prediction = lin.predict(new_data)

print(f"Predicted Daily Revenue: ${prediction[0]:,.2f}\n",'='*100)
```

```
Predicted Daily Revenue: $8,955.44
 ==========================================================================================
===============
```